Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

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A fixed-width language-model representation may support a semantic-feature dictionary far larger than its exact carrier dimension. Let d be the rank of a declared real carrier, fix an absolute-inner-product tolerance $0 < \varepsilon < 1$, and let $M_\varepsilon(d)$ be the maximum cardinality of a unit-vector family in $\mathbb{R}^d$ whose absolute pairwise inner products do not exceed ε. Lévy concentration explains why almost every direction is nearly orthogonal to any fixed direction. The Johnson–Lindenstrauss lemma gives the complementary finite-set statement: logarithmic dimension suffices to preserve a prescribed finite geometry, and its standard-basis construction gives the existence bound $M_\varepsilon(d) \geq \exp(c\varepsilon^2 d)$ for fixed ε and sufficiently large d. Direct random spherical coding gives the same exponential scale, while the Welch bound supplies a deterministic coherence floor. If an independently justified full tensor-product carrier has $d = q^n$, with fixed $q > 1$, this becomes $M_\varepsilon(q^n) \geq \exp(c\varepsilon^2 q^n)$, doubly exponential in the factor count n. For an ordinary Transformer residual stream, however, only this ambient-code existence result in measured d follows; whether a learned dictionary approaches it is empirical, and tensor-product growth is an additional hypothesis.

Geometric coexistence is not the same as simultaneous semantic access. For k random-like active directions, a fixed-target correlation readout has interference of order $\sqrt{k/d}$. More specifically, learning should place smaller overlaps between important features that frequently coactivate, reducing coactivation-weighted interference relative to isotropic and activity-matched controls. On this classical geometry we formulate a separate model of lexical ambiguity: context operators share a token-associated invariant sector and differ on sense subspaces in its orthogonal complement. Under spectral separation, the corresponding observables share the projector onto that sector as a spectral projector. A learned non-oracular router selects or mixes the associated linear filters. The model is quantum-inspired rather than physically quantum, and its content is the testable advantage of this shared-sector constraint over parameter-matched classical alternatives on held-out data.

I. INTRODUCTION

This chapter examines contextual representation in large language models (LLMs) as part of the volume's broader inquiry into "quantum thinking." Only a minimal orientation to LLM operation is needed here. Input text is first divided into tokens, and each token is mapped to an initial vector. Transformer blocks then use self-attention and feed-forward transformations to turn those initially context-free vectors into sequence-dependent hidden states [1]. An output layer converts the final state into token scores, and a softmax distribution supports probabilistic next-token selection. This compact chain—tokens, embeddings, contextual transformation, and finite readout—supplies the computational background for the argument below. Section II expands it in plain language and distinguishes persistent model weights, transient attention coefficients, contextual hidden states, and optional online updates. Appendix A gives the corresponding minimal formalization.

Current Transformer-based LLMs are classical computational systems. They run on classical hardware and use real-valued matrix operations, normalization, nonlinearities, copying, and generally irreversible updates. Their reliance on vectors, inner products, high dimension, or probabilistic output does not by itself make them quantum or even quantum-like. The more limited question pursued here is where ordinary high-dimensional geometry already explains the coexistence and contextual selection of semantic features, and where a stronger comparison with quantum-like formalisms might begin. Such a comparison would have to do more than rename familiar neural operations: it would need to identify a role for contextual bases or subspaces, incompatible or noncommuting alternatives, interference, or measurement-like readout that yields explanatory or predictive value beyond a classical dynamic-embedding account.

Semantic concurrency makes this boundary concrete. A model must preserve many latent possibilities while making only a small, context-selected subset operationally accessible. Throughout this chapter, *semantic expressibility* means the number and range of task-relevant distinctions that remain recoverable under a declared feature extractor, activity distribution, decoder, and error tolerance. A large collection of vectors alone is not semantic expressibility.

Lexical ambiguity is the simplest illustration: the same token can participate in financial, river-edge, or seat/bench uses of "bank," yet the surrounding sequence drives it toward different contextual representations. The umbrella includes both related polysemous senses and historically distinct homonymous uses [2]; the formal construction does not depend on that lexicographic

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classification. This relation among possibility, context, and readout connects directly to the volume's recurring themes without implying that the underlying machine is physically quantum. It also motivates a separation between what a representation can geometrically accommodate and what a finite channel can reliably recover.

The argument has three evidential levels. First, Lévy concentration, Johnson–Lindenstrauss (JL) dimension reduction, spherical coding, and the Welch bound are established mathematics. Here they supply complementary accounts of quasi-dimensionality, not new theorems. Second, their application to a learned semantic dictionary is an empirical model whose carrier, metric, feature extractor, activity distribution, and decoder must be specified. Third, the shared-sector organization of lexical ambiguity is a stronger hypothesis tested against classical controls. No physical quantumness, Born-rule dynamics, or semantic Kochen–Specker theorem is inferred.

Let d be the exact rank of a declared usable carrier, and let M be the number of distinct unit feature directions returned by a prespecified extraction and duplicate-removal rule. Only d is a vector-space dimension; M is an empirical dictionary cardinality. For a fixed amplitude tolerance $0 < \varepsilon < 1$, call a family quasi-orthogonal when all absolute pairwise inner products are at most ε. Lévy concentration gives the typicality statement

$$\mathbb{P}\{|u^{\mathsf{T}}x| > \varepsilon\} \leq 2\exp(-cd\varepsilon^2) \tag{1}$$

for a fixed unit u and an isotropic random unit x. JL gives a complementary finite-configuration construction which, after converting distance distortion to coherence, yields

$$M_\varepsilon(d) \geq \exp(c'\varepsilon^2 d), \tag{2}$$

where $M_\varepsilon(d)$ is the maximum achievable code cardinality under this condition. This is an existence result for fixed ε and sufficiently large d, not a claim that the empirical dictionary size M attains the bound.

The double exponential requires a second, independent amplification. If a carrier has the justified full tensor-product structure $\bigotimes_{a=1}^{n} \mathbb{R}^q$, with fixed $q > 1$, so that $d = q^n$, then Eq. (2) becomes

$$M_\varepsilon(q^n) \geq \exp\bigl(c'\varepsilon^2 q^n\bigr), \tag{3}$$

doubly exponential in n for fixed $0 < \varepsilon < 1$. This is a two-stage amplification: composition first makes the exact carrier exponential, and quasi-orthogonal coding is exponential in that carrier. A standard Transformer residual stream is supplied directly as $\mathbb{R}^d$ and is not thereby a full tensor product of semantic subsystems. For LLMs the robust conclusion is therefore an ambient-code existence lower bound exponential in measured d; the double exponential is a conditional tensor-product hypothesis.

Abundance also creates crowding. Among M independently sampled isotropic directions, the maximum of the $\binom{M}{2}$ pairwise overlaps is typically $2\sqrt{\log M/d}$ in the small-overlap regime. More importantly, coexistence does not guarantee simultaneous access. If k random-like directions are active with order-one amplitudes, a fixed-target correlation readout accumulates interference

$$\sigma_{\text{int}} \sim \sqrt{\frac{k}{d}}. \tag{4}$$

The semantic prediction is that learning places smaller overlaps where they are operationally expensive: between important features that frequently coactivate. Coactivation-weighted interference is therefore compared with isotropic and activity-matched controls.

The chapter finally advances an independent hypothesis about lexical ambiguity. For some tokens with stable sense inventories, context may be represented by symmetric operators that share a stable token-associated invariant sector while differing on low-dimensional sense subspaces in its orthogonal complement. A learned router selects or mixes their projections. Concentration geometry does not imply this structure; it earns scientific content only if it improves held-out prediction or compression over parameter-matched unstructured models. Section VII C states how every layer of the proposal can fail.

A serious caveat is anisotropy. The nominal width of an LLM layer may exceed the dimension of the carrier actually occupied by its states. Throughout this chapter d therefore means the *usable exact dimension*: first restrict or whiten the representation to the carrier being tested, then let d be the size of an orthonormal basis of that carrier. If this usable d is small, the random-code crowding and readout baselines worsen. The method used to identify the carrier must therefore be declared empirically.

The reference-subspace and operator hypotheses have a narrower domain. They are meant for lexically ambiguous tokens with established WordNet- or OntoNotes-style sense inventories and enough SemCor- or WiC-style contextual evaluation data [3]. Failure on a representative sample, against the controls in Sec. VII C, would count against a privileged token reference subspace or shared-sector operator model even if the broader null calibration survives.

The proposal is adjacent to the superposition hypothesis in mechanistic interpretability [4], according to which neural networks represent more features than they have neurons by packing features into nearly orthogonal directions. The present chapter separates the geometric background from a possible semantic organization within it. The general superposition picture concerns an overcomplete dictionary of feature directions and sparse active sets. The present model adds the more specific possibility that contextual readout is organized around stable token-associated directions and context-dependent, potentially noncommuting projectors.

Section II separates creation of the trained weights, autoregressive response generation, and possible recursive online adaptation. Section III develops the geometric null model through Lévy concentration, Johnson–Lindenstrauss dimension reduction, direct random coding, and the deterministic Welch floor. Section IV separates dictionary coexistence from simultaneous readout and culminates in the task-dependent criterion of Sec. IV C. Section V introduces the reference-subspace hypothesis, its shared-sector projector and observable parameterization, and its relation to dynamic contextual embeddings. Section VI connects the construction to attention as soft context routing rather than literal basis reconstruction. Section VII discusses superposition, sparse autoencoders, limitations, empirical tests, and the limited quantum analogy.

Global variables and local ranks. Throughout, $\mathbb{R}^d$ is the usable Euclidean carrier, and d is its exact integer dimension. The symbol M is the size of a prespecified semantic feature dictionary; it is a count of directions, not another dimension. A dictionary is called overcomplete when $M > d$, and its directions are called ε-quasi-orthogonal only if their measured overlaps satisfy the criterion below. The function $M_\varepsilon(d)$ denotes a tolerance-dependent code cardinality, not a new dimension. The symbol k counts directions active in one context. The optional n counts local factors only when a full tensor-product structure $d = q^n$ has been independently specified. The quantities ε and the later rms-interference tolerance Δ are not dimensions, while p and m_s are local ranks of particular reference and sense subspaces. None is a second global vector-space dimension. Absolute inner products provide a conservative, sign-insensitive coherence diagnostic. When feature orientation carries semantic information, the signed overlap distribution must be reported as well; no claim that all semantic features are projective rays is assumed.

The carrier has inner product $u^\mathsf{T} v$ and norm $\|v\| = \sqrt{v^\mathsf{T} v}$; $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$ is the unit sphere; I is the identity matrix; $\|\cdot\|_{\mathrm{op}}$ is the operator norm; and $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the uniform measure on the relevant sphere unless stated otherwise. Universal positive constants are denoted c, c', possibly changing from line to line.

II. FROM TEXT TO WEIGHTS AND BACK TO TEXT: THREE COMPUTATIONAL LOOPS

The word *weight* is used for two different objects in informal accounts of Transformers. A trained model has persistent numerical parameters: its embedding table, query, key, value, output, and feed-forward matrices, normalization parameters, and final vocabulary readout. Self-attention also computes input-dependent coefficients that are often called attention weights. Those coefficients exist only for the current forward pass. They do not become persistent model parameters. Keeping this distinction visible separates the three computational loops below.

A. Creating the weights: tokenization, contextual vectors, and training

Training begins with text, but a neural network does not receive words or sentences as primitive objects. A tokenizer first segments character strings into a finite vocabulary of token types and replaces each occurrence by an integer identifier. Tokens may be whole words, word fragments, punctuation, spaces, or byte-level units. Modern subword schemes are designed to cover rare and previously unseen words without assigning a separate vocabulary entry to every possible word [5, 6]. The tokenizer is normally learned or chosen before the main language-model optimization and is then held fixed during that optimization.

An embedding table associates each token identifier with an initial real vector. At this first lookup stage, two occurrences of the same token use the same table entry. Position information is also supplied, either through added position vectors or through mechanisms such as rotary transformations inside attention; some systems add role or segment information as well. The lookup vector is therefore not yet the fully contextual meaning of that occurrence. Context dependence is created by the successive Transformer layers.

Within one self-attention head, the current vector at each position is mapped by learned matrices to a query, a key, and a value. The query at one position is compared by an inner product with keys at other permitted positions. In the decoder-only autoregressive Transformer considered here, a causal mask excludes future positions during next-token training. Softmax normalization turns the permitted comparisons into nonnegative attention coefficients, and their weighted sum of the value vectors supplies a context-dependent update. Multiple heads can learn different patterns of dependence. Residual connections, normalization, and feed-forward blocks then transform the update further. After many layers, the hidden vector for an occurrence of "bank" can differ substantially between financial, river-edge, and seat/bench contexts even though all occurrences began from the same token-table entry [1].

This forward calculation still does not create new persistent weights. The last hidden state at each training position is converted into scores for the next token. The discrepancy between the predicted distribution and the observed next token defines a training loss. Backpropagation differentiates that loss through the output map, feed-forward blocks, attention operations, and embedding table. An optimizer then changes the persistent parameters. Repeating this cycle over many batches creates the trained weight values. In short, self-attention produces contextual activations and differentiable routes through which error signals flow; gradient-based optimization produces the persistent model weights [7].

Does this description require a Hilbert space? Nothing here requires more than finite-dimensional Euclidean inner-product geometry. Because $\mathbb{R}^d$ is complete, it may formally be called a real Hilbert space, but completeness plays no computational role and implies no quantum mechanism. The construction needs ordinary real vectors, inner products, matrices, and nonlinear maps; it does not require complex amplitudes, Born probabilities, or quantum hardware. Nor does training assign one exactly orthogonal vector to every token or every meaning. The quasi-orthogonal geometry studied later concerns a declared dictionary of extracted feature directions at a chosen resolution. It is a possible organization of learned semantic distinctions, not an assumption built into tokenization or self-attention.

B. Generating an output: tokenization and autoregressive feedback

Response generation begins with the same tokenizer. The prompt, including any control, role, or formatting tokens exposed to the model, is converted to a token sequence and then to initial vectors. The trained parameters are now normally fixed. A causal forward pass produces contextual hidden states for the prompt, and the final relevant state is mapped to one score for every token in the vocabulary. Softmax converts these scores into a next-token distribution. Greedy decoding chooses its largest entry; sampling may use a temperature or restrict attention to a high-probability subset before drawing a token.

The selected token is appended to the sequence. The enlarged sequence is then the context for the following prediction, so the generated token can change all subsequent attention patterns and hidden states. This loop is repeated until an end condition is reached. Implementations commonly cache previous keys and values to avoid recomputing unchanged parts of the prefix, but caching changes efficiency rather than the trained parameters. Finally, the tokenizer's inverse assembly rules turn the generated token identifiers back into a character string [6].

Autoregressive generation is thus already recursive in a precise but limited sense: each output token becomes part of the next input. The context, attention coefficients, and hidden vectors evolve from step to step, while the persistent model weights remain fixed. The standard decoding interface does not require a separately represented complete answer; it emits tokens successively by repeatedly mapping the current vector-valued context to a distribution over the next discrete token.

C. Recursive adaptation, alignment, and hallucination

Three forms of recursion should be separated. The first is the autoregressive token feedback just described. The second adds an external memory, retrieval result, tool output, or longer conversation history to the next context without changing the model's persistent weights. The third is genuine online learning: after an interaction, a loss or verified feedback signal updates all parameters, an adapter, or a set of designated fast weights. Dynamic evaluation and test-time training show that this third loop is technically possible [8]. It is not, however, the ordinary inference rule of the Transformer considered here, and it is not normally an unrestricted update after every dialogue.

Alignment and safety are among the reasons to constrain such online updates, but they are not the only reasons. A user interaction usually supplies no verified target against which a gradient should be computed. Unrestricted updates can cause catastrophic forgetting, distributional drift, loss of reproducibility, privacy leakage, or deliberate data poisoning through prompt injection. Computation and rollback are additional concerns. A defensible online learner therefore needs provenance, external verification, bounded or adapter-local updates, held-out tests, and a recoverable reference model. Recursive training on unverified model output can instead reinforce errors and eventually degrade the output distribution [9].

The base model produced by next-token pre-training approximates regularities in its corpus. Supervised instruction tuning and preference-based methods then deliberately move the parameters or the effective inference policy toward chosen notions of helpfulness, harmlessness, instruction following, and safety [10]. Relative to parameters favored by the raw next-token objective, this is deliberately an *objective tilt*. Calling it a geometric distortion would additionally require a declared parameter-space or output-distribution metric. In either vocabulary, the change alone does not imply that factual performance has deteriorated: the purpose of post-training is to replace the raw objective by a more useful one.

There is nevertheless a concrete version of the concern that alignment can contribute to hallucination. If human or learned preferences reward fluent, confident, accommodating answers more strongly than calibrated abstention, optimization can favor agreement or plausible completion over epistemic caution. Preference judgments have been shown to contribute to sycophantic behavior in some settings [11]. The same pressure could increase unsupported claims. Conversely, carefully constructed human feedback has also improved truthfulness in matched evaluations, and hallucination already arises in base-model prediction without alignment post-training [10, 12]. Alignment and factual accuracy are therefore distinct axes: either can improve while the other worsens.

The claim that alignment contributes greatly to hallucination should thus be posed as a causal, model- and protocol-specific hypothesis. Starting from the same initialization or base checkpoint, one should compare declared post-training objectives with matched training data, optimization and compute budgets, and multiple matched seeds. Evaluation should use the same prompt distribution, grounding information, and decoding rule and should measure unsupported assertions, calibration, coverage, and abstention. The sign and magnitude of the change are empirical. Appendix A 3 states this test without building the desired conclusion into the notation.

III. QUASI-DIMENSIONALITY FROM LÉVY CONCENTRATION AND JOHNSON–LINDENSTRAUSS

The nontrivial question is how the cardinality of a finite-resolution code scales with its exact carrier dimension. Here d is the usable carrier rank, whereas M counts the directions in one measured dictionary and may exceed d. Let

$$W = \{w_1, \dots, w_M\} \subset S^{d-1}, \qquad \mu(W) = \max_{i \neq j} |w_i^\mathsf{T} w_j|. \tag{5}$$

For $0 < \varepsilon < 1$, we call this particular dictionary ε-quasi-orthogonal when $\mu(W) \le \varepsilon$, and write $M_\varepsilon(d)$ for the maximum cardinality achievable in $\mathbb{R}^d$ under this condition. Thus $M_\varepsilon(d)$ and the empirical count M are cardinalities, not additional vector-space dimensions. The term *quasi-dimensionality* will refer only to this resolution-dependent abundance of directions.

The mathematical ingredients below are established results. Their purpose is to distinguish three claims that should not be conflated: Lévy concentration says that near-orthogonality is typical relative to a fixed direction; Johnson–Lindenstrauss (JL) gives a finite-set construction at logarithmic dimension; and direct random coding provides a null ensemble for the largest overlap in a dictionary. None of these facts alone identifies semantic features in an LLM.

A. Lévy concentration and angular homogenization

In one standard form of Lévy's lemma, if $f : S^{d-1} \to \mathbb{R}$ is L-Lipschitz in chordal distance and m_f is a median of f, then

$$\mathbb{P}\{|f - m_f| \ge t\} \le 2 \exp\left[-c\,\frac{(d-1)t^2}{L^2}\right] \tag{6}$$

for a universal constant $c > 0$ [13, 14]. The choice of median or mean and the numerical constant vary among formulations; the invariant content is exponential concentration in dimension for every regular scalar probe.

Fix $u \in S^{d-1}$ and take $f_u(x) = u^\mathsf{T} x$. This function is 1-Lipschitz and, by rotational symmetry, has median and mean zero. For uniform $x \in S^{d-1}$,

$$\mathbb{E}[u^\mathsf{T} x] = 0, \qquad \mathrm{Var}(u^\mathsf{T} x) = \frac{1}{d}, \tag{7}$$

and the corresponding spherical-cap estimate can be written

$$\mathbb{P}\{|u^\mathsf{T} x| \ge \varepsilon\} \le 2 \exp\left[-\frac{(d-1)\varepsilon^2}{2}\right], \qquad 0 < \varepsilon < 1. \tag{8}$$

Thus almost every direction lies close to the equator perpendicular to a specified direction, with a typical overlap amplitude of order $d^{-1/2}$.

There is an equivalent angular description. If the sign of a direction is irrelevant, its principal angle from u is $\theta_{\mathrm{pr}} = \arccos |u^\mathsf{T} x|$, and

$$\theta_{\mathrm{pr}} = \frac{\pi}{2} - O_{\mathbb{P}}(d^{-1/2}). \tag{9}$$

In the independently sampled isotropic null, a randomly chosen pair therefore exhibits *angular homogenization*: its principal angle lies in a band of width $O_{\mathbb{P}}(d^{-1/2})$ near $\pi/2$. Increasing the dictionary size at fixed d does not narrow this typical-pair distribution; instead it increases the extreme overlap, as quantified below. High dimension creates both the typical angular concentration and room for exponentially large finite codes.

For comparison with the quantum-inspired formalism below, the corresponding complex statement is especially sharp. If ϕ, ψ are independent Haar-random unit vectors in $\mathbb{C}^d$, then $Z = |\langle \phi, \psi \rangle|^2$ has the $\mathrm{Beta}(1, d-1)$ distribution. Using the same ε for an overlap *amplitude*, rather than changing midway to a squared-overlap tolerance, gives the exact tail

$$\mathbb{P}\{|\langle \phi, \psi \rangle| \ge \varepsilon\} = (1 - \varepsilon^2)^{d-1} \le \exp[-(d-1)\varepsilon^2]. \tag{10}$$

The real and complex formulas differ in constants and exact distributions, but both exhibit the same operative exponential scale $\exp[-cd\varepsilon^2]$.

B. Johnson–Lindenstrauss and a constructive exponential code

Lévy's lemma is a measure statement on a sphere. JL is complementary: it preserves a specified finite metric configuration. For $0 < \delta < 1$ and a finite set X, there is a linear map Φ into $\mathbb{R}^d$ such that

$$\begin{aligned}(1-\delta)\|x - y\|^2 &\le \|\Phi x - \Phi y\|^2 \\ &\le (1+\delta)\|x - y\|^2, \qquad x, y \in X,\end{aligned} \tag{11}$$

provided

$$d \ge C_{\mathrm{JL}} \delta^{-2} \log |X| \tag{12}$$

for a universal C_{JL} [15, 16].

To make the connection to quasi-orthogonality explicit, take

$$X = \{0, e_1, \dots, e_M\} \subset \mathbb{R}^M.$$

Including the origin is important: Eq. (11) then controls both the norms $\|\Phi e_i\|$ and the distances $\|\Phi e_i - \Phi e_j\|$. The real polarization identity gives

$$2(\Phi e_i)^\mathsf{T}(\Phi e_j) = \|\Phi e_i\|^2 + \|\Phi e_j\|^2 - \|\Phi e_i - \Phi e_j\|^2. \tag{13}$$

Consequently $|(\Phi e_i)^\mathsf{T}(\Phi e_j)| \le 2\delta$, while $\|\Phi e_i\|^2 \ge 1-\delta$. After normalization, $w_i = \Phi e_i/\|\Phi e_i\|$, one obtains

$$|w_i^\mathsf{T} w_j| \le \frac{2\delta}{1-\delta} \qquad (i \neq j). \tag{14}$$

Taking $\delta = \varepsilon/(2+\varepsilon)$ makes the right-hand side of Eq. (14) equal to ε. Reading Eq. (12) constructively therefore gives, for every fixed $0 < \varepsilon < 1$ and all sufficiently large d,

$$M_\varepsilon(d) \ge \exp(c'\varepsilon^2 d) \tag{15}$$

for a universal $c' > 0$. This is an existence lower bound. Sharp lower bounds for worst-case JL dimension reduction [17, 18] do not, by themselves, establish a matching universal upper bound for spherical codes, and no such upper bound is claimed here.

C. Direct random coding, the Welch floor, and two amplifications

The exponential scale can also be reached directly. Draw the M directions independently from the uniform measure on S^{d-1}. Applying Eq. (8) to all unordered pairs gives

$$\mathbb{P}\{\mu(W) \ge \varepsilon\} \le M(M-1)\exp\left[-\frac{(d-1)\varepsilon^2}{2}\right]. \tag{16}$$

Whenever the right-hand side is less than one, at least one ε-quasi-orthogonal dictionary exists. This reproduces the $\exp(c\varepsilon^2 d)$ lower scale of the JL construction by a different route. Equation (16) is only a sufficient random-code bound; it is neither an optimized-code upper bound nor a semantic capacity theorem.

Equating the number of pairs to the real spherical-cap tail gives the useful finite-sample null estimate

$$\mu_{\max}^{\text{null}}(d, M) \approx 2\sqrt{\frac{\log M}{d}}, \qquad \log M \ll d, \tag{17}$$

up to lower-order spherical-cap factors. It describes the most extreme pair in an independently sampled isotropic dictionary, not the overlap of a typical pair and not a hard limit on a learned one.

A genuine deterministic restriction is supplied by the Welch bound:

$$\mu(W)^2 \ge \frac{M-d}{d(M-1)} \qquad (M > d) \tag{18}$$

for every unit-vector dictionary [19]. The Welch floor, the random maximum in Eq. (17), and the existence lower bound in Eq. (15) answer different questions and should not be presented as the two edges of one universal interval.

The exact complex-Haar law in Eq. (10) makes the corresponding random-code statement particularly transparent. If $M_\varepsilon^{\mathbb{C}}(d)$ denotes the maximum cardinality in $\mathbb{C}^d$ at the same amplitude tolerance, a union bound over all pairs gives the explicit existence result

$$M_\varepsilon^{\mathbb{C}}(d) \ge \left\lfloor \exp\left[\frac{d-1}{2}\left(-\log(1-\varepsilon^2)\right)\right]\right\rfloor. \tag{19}$$

This is the sharper complex analogue of the real exponential scale, not an additional linear dimension.

The conditional double exponential requires two separate amplifications. If an independently justified carrier is the full tensor product $\bigotimes_{a=1}^n \mathbb{R}^q$, with fixed $q > 1$, then its exact dimension is

$$d = q^n. \tag{20}$$

For fixed $0 < \varepsilon < 1$, substitution into Eq. (15) gives

$$M_\varepsilon(q^n) \ge \exp\bigl(c'\varepsilon^2 q^n\bigr), \tag{21}$$

which is doubly exponential in n. Equation (19) gives the same conclusion, with an explicit exponent, for a full complex tensor-product carrier. These bounds concern arbitrary directions in the full carrier. They do not follow if admissible states are restricted to product vectors or another low-complexity manifold; access to the full carrier, or a code construction within the admissible class, is an additional premise. An ordinary Transformer residual stream is presented only as a measured real carrier $\mathbb{R}^d$. Its width, layers, heads, or token positions do not by themselves establish independent semantic tensor factors. For an LLM the unconditional geometric statement is therefore an existence lower bound exponential in the declared usable d; Eq. (21) is a further tensor-product hypothesis, not an architectural consequence.

IV. FROM GEOMETRIC COEXISTENCE TO SEMANTIC READOUT

An exponential dictionary can coexist geometrically without all its coefficients being simultaneously accessible. Readout depends on which directions activate together, their amplitudes, the decoder, and the required error probability. This distinction is where the geometry acquires potential semantic content.

A. Fixed-target, worst-case, and joint readout

Let an active set S of size k produce the hidden state

$$h = \sum_{j\in S} x_j w_j. \tag{22}$$

For one specified active target $i \in S$, a correlation decoder returns

$$w_i^\mathsf{T} h = x_i + \sum_{\substack{j\in S\\ j\neq i}} x_j w_i^\mathsf{T} w_j. \tag{23}$$

In the exact independently sampled isotropic ensemble, provided S and the amplitudes are fixed or selected independently of the direction draws, conditional on S, (x_j), and w_i, the variables $w_i^\top w_j$ for $j \neq i$ are independent, centered, and have variance $1/d$. Consequently

$$\operatorname{Var}\left[\sum_{\substack{j\in S\\ j\neq i}} x_j w_i^\top w_j \;\middle|\; S,(x_j),w_i\right] = \frac{1}{d}\sum_{\substack{j\in S\\ j\neq i}} x_j^2. \tag{24}$$

For a learned, merely random-like dictionary this equality is used only as an approximate null calibration. For comparable order-one amplitudes it gives the fixed-target rms scale

$$\sigma_{\text{int}} \sim \sqrt{\frac{k-1}{d}} \simeq \sqrt{\frac{k}{d}}. \tag{25}$$

This estimate is a property of the declared correlation decoder under cancellation assumptions, not a universal recovery law.

Without cancellation, coherence yields the deterministic target-wise bound

$$|w_i^\top h - x_i| \le \mu(W) \sum_{\substack{j\in S\\ j\neq i}} |x_j|. \tag{26}$$

The distinction becomes sharper for joint recovery. With W_S the matrix whose columns are the active directions and $G_S = W_S^\top W_S$, simultaneous correlation readout gives

$$\widehat{x}_S = W_S^\top h = G_S x_S, \qquad \|G_S - I\|_{\text{op}} \le (k-1)\mu(W), \tag{27}$$

where the last inequality follows, conservatively, from Gershgorin's theorem. Support identification must additionally control false readouts among the $M-k$ inactive directions, introducing a multiple-comparison cost. Uniform sparse recovery therefore needs stronger hypotheses. For suitable random subgaussian measurement dictionaries, a representative sufficient restricted-isometry scale is $d \gtrsim C_{\text{RIP}} k \log(M/k)$ [20–22]; this scale is not implied for an arbitrary W by the coherence or JL statements above. The exponential coexistence of directions, fixed-target rms access, worst-case access, and joint coefficient recovery are four distinct statements.

B. Why semantic interference must be conditioned on coactivation

Maximum coherence assigns the same cost to every pair. A semantic decoder does not: two directions that never activate together can overlap without contributing to the error in Eq. (23). Conversely, even a moderate overlap can be expensive when the pair coactivates often, carries large amplitudes, and matters to the task. Define the ordered, nonnegative activity-and-importance weight

$$a_{ij} := \omega_i \, \mathbb{E}\left[\mathbf{1}_{\{i,j \text{ active}\}} x_j^2\right], \qquad \omega_i \ge 0, \tag{28}$$

where ω_i is a nonnegative scalar recording the task importance of accurately reading feature i. If, for each target i, interference contributions from distinct j are uncorrelated on the held-out distribution, the diagonal part of the expected importance-weighted squared error is

$$E_{\text{int}} = \sum_{i\neq j} a_{ij} (w_i^\top w_j)^2. \tag{29}$$

We use E_{int} as a coactivation-weighted diagonal interference proxy. Without the uncorrelated-contribution assumption, the full squared error also contains cross-covariance terms involving distinct j and j'.

This changes the scientific prediction. A useful learned dictionary need not beat the isotropic null on every pair, and a nonzero overlap is not itself a defect. If training is pressured to reduce semantic readout error, then important, frequently coactive pairs should contribute less to Eq. (29) than they do in controls matched for d, M, marginal activity, amplitude, and feature importance. That conditional redistribution, not overcompleteness alone, is testable.

C. A task-dependent operating window

There is no task-independent numerical bracket whose lower edge is d and whose upper edge is a universal packing capacity. A meaningful operating window is instead the intersection of an empirical coverage requirement with declared readout tolerances. Let $\varepsilon_{\text{task}}$ be the largest tolerated absolute pairwise overlap for a chosen diagnostic and let Δ be the tolerated fixed-target rms interference. For the real isotropic null, Eqs. (17) and (25) give the separate crossover scales

$$M_{\text{crowd}}(d,\varepsilon_{\text{task}}) \approx \exp\left(\frac{d\varepsilon_{\text{task}}^2}{4}\right), \qquad k_{\text{rms}}(d,\Delta) \lesssim d\Delta^2. \tag{30}$$

The first concerns the worst of order M^2 pairs formed from M independently sampled directions; the second concerns one specified target among k active directions. Let $M_{\text{req}}(\mathcal{T})$ be the smallest measured dictionary size that reaches a prespecified coverage or performance level on task $\mathcal{T}$, and let $E_{\max}(\mathcal{T})$ be its allowed value of the interference proxy. A useful task-conditioned summary is

$$\begin{aligned} M_{\text{req}}(\mathcal{T}) \le M &\lesssim M_{\text{crowd}}(d,\varepsilon_{\text{task}}),\\ k \lesssim k_{\text{rms}}(d,\Delta), &\qquad E_{\text{int}} \le E_{\max}(\mathcal{T}). \end{aligned} \tag{31}$$

The upper relation for M is an isotropic-null design heuristic, not a universal spherical-code upper bound. A stricter joint-recovery criterion may replace it through Eq. (27). The Goldilocks region is thus the set of dictionaries that provide enough measured task coverage while satisfying explicitly selected pairwise, active-set, and readout-error criteria. Its lower edge comes from the task's coverage curve, not from an orthogonality count.

For scale, if $d = 4096$ and $M = 10^4$, Eq. (17) predicts a random-dictionary maximum of about 9.5%, although one typical pair has rms overlap only $1/\sqrt{d} \approx 1.6\%$. Separately, $k = 64$ gives the fixed-target scale $\sqrt{k/d} = 0.125$. These numbers illustrate why the dictionary count, the most extreme pair, and simultaneous readout cannot be collapsed into one effective dimension.

Nothing in these benchmarks is specifically semantic until a feature extractor, activity distribution, decoder, and task are fixed. The next section adds a separate and stronger hypothesis about how context-dependent readout may be organized for lexically ambiguous tokens.

V. TOKEN REFERENCE SUBSPACES AND SHARED-SECTOR CONTEXT OPERATORS

The previous section supplied classical geometric benchmarks under declared assumptions; it did not imply any particular organization of lexical ambiguity. We now ask a separate, more speculative question: does a trained model organize the contextual states of some lexically ambiguous tokens relative to a stable token-associated reference subspace? The hypothesis is scientifically useful only if that constraint generalizes to held-out data better than equally expressive unstructured alternatives.

This is also the bridge from global quasi-dimensionality to local meaning. Lévy concentration and JL concern how many candidate directions can coexist at a declared angular resolution. A linguistic context does not activate or decode that entire repertoire. It selects a small active set or low-dimensional subspace in which the distinctions relevant to the present token must remain accessible. The construction below proposes one particular organization of that selection: a token-associated sector is shared, while class-specific sense content lies in transverse subspaces. The global exponential code size and this local context structure are complementary claims, not consequences of one another.

Fix one layer ℓ and one centered or whitened carrier for each test. The observed sequence context is x, while s denotes only the discrete operator class that indexes a candidate filter. In supervised experiments an operator class may be aligned with a labeled word sense; in unsupervised experiments it is latent. We abbreviate $h_\ell(t \mid x)$ by $h(x)$ when t and ℓ are fixed. The notation distinguishes mathematical types and roles:

- $r_t \in \mathbb{R}^d$ is a unit candidate token reference vector;
- $\mathcal{R}_t \subset \mathbb{R}^d$ is the token reference subspace;
- P_t is the orthogonal projector onto $\mathcal{R}_t$;
- $S_{t,s} \subset \mathcal{R}_t^\perp$ is a low-dimensional sense subspace;
- $u_{t,s,j} \in S_{t,s}$ are vectors spanning that sense subspace;
- $h_\ell(t \mid x) \in \mathbb{R}^d$ is an observed contextual hidden state.

Thus r_t, $u_{t,s,j}$, and $h_\ell(t \mid x)$ are vectors, whereas $\mathcal{R}_t$ and $S_{t,s}$ are subspaces. The w_i used in Sec. III have a separate role: they are generic feature directions in the geometric null model and are not reused as sense labels.

A. The reference-subspace hypothesis

In the simplest version, let r_t be a candidate unit direction and set

$$\mathcal{R}_t = \mathrm{span}\{r_t\}. \tag{32}$$

Empirically, r_t could be a normalized mean contextual embedding, a learned token-specific probe direction, or a static input embedding after an explicit map into the same layer carrier. It is simply a hypothesized context-stable reference for token identity. It is not a mechanical or dynamical object. Any state can be decomposed relative to any direction, so the decomposition alone has no empirical content. The claim is that a token-associated $\mathcal{R}_t$ exposes sense structure better than matched control directions or subspaces.

For each operator class s, define a low-dimensional sense subspace

$$\begin{aligned} &S_{t,s} = \mathrm{span}\{u_{t,s,1}, \ldots, u_{t,s,m_s}\}, \\ &S_{t,s} \subset \mathcal{R}_t^\perp, \qquad m_s \ll d - \dim(\mathcal{R}_t). \end{aligned} \tag{33}$$

For the intended three-arm example, finance, river-edge, and seat/bench associations of "bank" correspond to different subspaces $S_{\mathrm{bank},s}$ inside the common orthogonal carrier $\mathcal{R}_{\mathrm{bank}}^\perp$. The verbal use in which an aircraft banks or tilts supplies a fourth sense and would add a fourth sense subspace; it is mentioned here but omitted from the drawing for clarity. A word sense is not a single vector: topic, syntax, selectional preferences, and world knowledge may require several spanning directions $u_{t,s,j}$.

The reference object may itself require more than one direction. In that case define

$$\mathcal{R}_t = \mathrm{span}\{r_{t,1}, \ldots, r_{t,p}\}, \qquad p \geq 1, \tag{34}$$

and require

$$S_{t,s} \subset \mathcal{R}_t^\perp. \tag{35}$$

Equation (32) is the special case $p = 1$. The bare subspace construction specifies only a pattern of sharing. In quantum logic, different contexts can share one or more observables or spectral projections; finite-dimensional examples with two shared observables are known [23]. In the semantic model, $\mathcal{R}_t$ is a common token-associated reference subspace, while the sense-specific subspaces lie in its orthogonal complement. These incidence relations alone imply neither noncommutativity nor quantum probability. The next subsection makes the stronger, optional operator structure explicit.

The low-rank assumption is empirical. It predicts that dominant within-sense variation is low-dimensional after projection onto $\mathcal{R}_t^{\perp}$. This is compatible with evidence that contextualized embeddings can separate word senses, but it is stronger: it asks whether such separation is privileged relative to a token-associated reference subspace [24, 25].

B. Shared-sector class projectors and observables

The shared-subspace hypothesis can be parameterized by a constrained family of class-conditioned projectors. Let $Q_t \in \mathbb{R}^{d\times p}$ have orthonormal columns spanning $\mathcal{R}_t$, so that $P_t = Q_t Q_t^{\mathsf{T}}$. For each class s, let $B_{t,s} \in \mathbb{R}^{d\times m_s}$ have orthonormal columns spanning $S_{t,s}$. Here Q_t and $B_{t,s}$ are basis matrices, not additional semantic directions.

The projector onto the token-plus-sense subspace is

$$\Pi_{t,s} = P_t + B_{t,s}B_{t,s}^{\mathsf{T}}. \tag{36}$$

All $\Pi_{t,s}$ have the common invariant sector $\mathcal{R}_t$ and differ on their sense sectors. In particular,

$$[\Pi_{t,s}, \Pi_{t,s'}] = [B_{t,s}B_{t,s}^{\mathsf{T}}, B_{t,s'}B_{t,s'}^{\mathsf{T}}] \tag{37}$$

need not vanish. Crucially, the common term P_t cancels from this commutator; sharing a sector does not cause noncommutativity. Moreover, two generic fitted projectors will often fail to commute, so a nonzero commutator alone is not evidence for a quantum-inspired account. It becomes discriminating only if its size predicts an independently specified order-sensitive contextual task or improves held-out prediction relative to matched projector models.

A symmetric class observable with the same basis-independent shared sector can be written

$$A_{t,s} = \lambda_t P_t + B_{t,s}\Lambda_{t,s}B_{t,s}^{\mathsf{T}}, \tag{38}$$

where the scalar λ_t is shared across classes and the diagonal $\Lambda_{t,s}$ is class-specific. If P_t is to be the spectral projector for the shared eigenvalue, assume that λ_t is distinct from zero and from every diagonal entry of every $\Lambda_{t,s}$, as well as from the spectra of any added complementary terms. If $p > 1$, λ_t is deliberately degenerate within $\mathcal{R}_t$, so the observable identifies the shared subspace rather than an arbitrary basis within it. As displayed, $A_{t,s}$ acts as zero on the remaining orthogonal complement; additional orthogonal spectral terms may resolve that complement into further outcomes. The projector family is the empirical core of the proposal; the eigenvalues add scientific content only if they are prespecified or learned and improve a held-out observable outcome, rather than being assigned after the geometry is known.

The corresponding class-conditioned projection is concrete:

$$h_{t,s}^{(\Pi)}(x) = \Pi_{t,s}h(x) = P_t h(x) + B_{t,s}B_{t,s}^{\mathsf{T}}(I - P_t)h(x). \tag{39}$$

For $h(x) \neq 0$, make the observable operational through its normalized quadratic expectation and filtered state and filtered state

$$o_{t,s}(x) = \frac{h(x)^{\mathsf{T}}A_{t,s}h(x)}{\|h(x)\|^2}, \qquad h_{t,s}^{(A)}(x) = A_{t,s}h(x). \tag{40}$$

The router may use $o_{t,s}$ as a score, and a fitted module may use either $h_{t,s}^{(\Pi)}$ or $h_{t,s}^{(A)}$ as its class-conditioned representation. One may select a single s, or take a soft mixture of these outputs as in Sec. VI. Equations (36)–(40) define a proposed constrained module or probe; they do not claim that current LLMs already implement it. Quantum-logic discussions sometimes call contexts with shared outcomes *intertwining* or interlocking. Here "shared-sector" names that incidence pattern while avoiding confusion with the operator-theoretic assertion that an intertwiner X satisfies $AX = XB$. Everything here uses ordinary real matrices on classical hardware.

C. Projection energy and subspace readout

Let $h(x) \in \mathbb{R}^d$ be a contextual state and retain the orthogonal projector P_t defined above. The two components are

$$\begin{aligned} h(x) &= h_{\mathrm{ref}}(x) + h_{\perp}(x), \\ h_{\mathrm{ref}}(x) &= P_t h(x), \\ h_{\perp}(x) &= (I - P_t)h(x) \in \mathcal{R}_t^{\perp}. \end{aligned} \tag{41}$$

The hypothesis is that most sense-disambiguating variation is present in $h_{\perp}(x)$, not that the decomposition itself is special. A basis-invariant score can be expressed directly with the orthonormal matrices already defined:

$$\begin{aligned} e_t(x) &= \|Q_t^{\mathsf{T}}h(x)\|^2 = \|P_t h(x)\|^2, \\ \beta_{t,s}(x) &= B_{t,s}^{\mathsf{T}}h_{\perp}(x), \\ f_{t,s}(x) &= \|\beta_{t,s}(x)\|^2 = \|B_{t,s}B_{t,s}^{\mathsf{T}}h_{\perp}(x)\|^2. \end{aligned} \tag{42}$$

Here e_t measures energy in the candidate reference subspace and $f_{t,s}$ measures projection energy in the candidate sense subspace. Both definitions work for any reference rank p and are invariant under changes of orthonormal basis within either subspace. An illustrative projection-energy score is

$$\gamma_{t,s}(x) = f_{t,s}(x). \tag{43}$$

The selected sense label is

$$s^*(x) = \arg\max_s \gamma_{t,s}(x). \tag{44}$$

Multiplying every $\gamma_{t,s}$ by the same positive function of e_t would not change this ranking; reference energy can instead supply an independently calibrated engagement or abstention threshold. The score is an illustrative probe, not a derivation of how an LLM must disambiguate. At

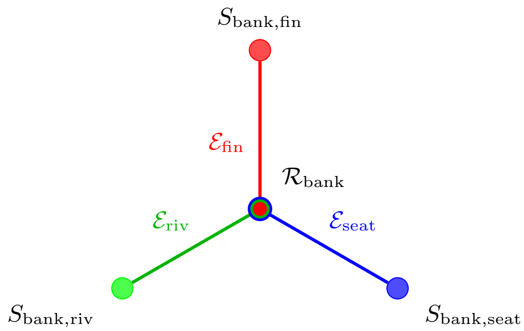

FIG. 1. Incidence schematic for the lexically ambiguous token "bank." The center is the single shared reference-subspace vertex $\mathcal{R}_{\text{bank}}$ and the outer nodes are the financial, river-edge, and seat/bench sense subspaces in $\mathcal{R}_{\text{bank}}^{\perp}$. Each colored arm denotes the displayed two-vertex fragment $\mathcal{E}_s = \{\mathcal{R}_{\text{bank}}, S_{\text{bank},s}\}$ associated with $\Pi_{\text{bank},s}$; it is not a vector, map, or causal arrow. The concentric colors at the center mark membership of the same shared vertex in all three displayed context-hyperedge fragments. Additional outcome sectors would complete each measurement context in a full contextuality hypergraph.

test time the true sense label is unavailable, so a practical implementation must learn the router from context without oracle access. The empirical content is the proposed separation of roles: the reference component tracks token identity or engagement, while the orthogonal component contains sense-disambiguating information. If no token-associated $\mathcal{R}_t$ exposes sense structure better than centered, token-mean, frequency-controlled, layer-matched, and random controls, the hypothesis fails.

Figure 1 illustrates these objects as an abstract hypergraph incidence structure.

D. Relation to dynamic contextual embeddings

Transformers already handle lexical ambiguity dynamically. The hidden state $h_\ell(t \mid x)$ of token t at layer ℓ depends on context x. The reference-subspace and shared-sector operator hypotheses are not alternative hardware architectures; they are proposed organizations and readouts of these observed trajectories.

Given a candidate reference subspace $\mathcal{R}_t$ and its projector P_t, write

$$\begin{aligned} h_\ell(t \mid x) &= P_t h_\ell(t \mid x) + z_\ell(t,x), \\ z_\ell(t,x) &= (I - P_t) h_\ell(t \mid x) \in \mathcal{R}_t^{\perp}. \end{aligned} \tag{45}$$

The decomposition is automatic; the hypothesis is not. It predicts that $P_t h_\ell(t \mid x)$ tracks token identity or engagement, while $z_\ell(t,x)$ carries most sense-disambiguating information. Sense labels should be more separable in $\mathcal{R}_t^{\perp}$ than in the orthogonal complements of matched control subspaces. For each sense s, the projected states $z_\ell(t,x)$ should also concentrate near the low-dimensional subspace $S_{t,s} \subset \mathcal{R}_t^{\perp}$.

Thus the critical question is not whether contextual embeddings move; they do. It is whether their sense-relevant movement has a privileged organization relative to a token-associated reference subspace, and whether conditional application of $\Pi_{t,s}$ explains that movement better than matched unstructured projectors.

VI. ATTENTION AND LATENT CONTEXT SELECTION

A standard attention head computes

$$h' = \sum_j \alpha_j \nu_j, \qquad \alpha_j = \frac{\exp(q_{\text{att}}^{\mathsf{T}} k_{\text{att},j}/\sqrt{d_{\text{head}}})}{\sum_{j'} \exp(q_{\text{att}}^{\mathsf{T}} k_{\text{att},j'}/\sqrt{d_{\text{head}}})}. \tag{46}$$

$q_{\text{att}}, k_{\text{att},j} \in \mathbb{R}^{d_{\text{head}}}$ are the query and key vectors of that head, and ν_j is its value vector. The local head width d_{head} is an architectural quantity, not necessarily the selected carrier dimension d used in the geometric null model. Keys and values are produced by different learned maps, so value vectors are not generally coefficients of a dual basis for the key vectors. Attention is therefore not literal basis reconstruction.

The useful analogy is softer. Attention routes information: it amplifies some contextual directions and suppresses others. In the terminology of Sec. V, this resembles context-dependent subspace selection. A literal basis-reconstruction interpretation would require additional constraints, such as approximately Parseval keys and values implementing the corresponding dual reconstruction. We do not assume this.

A usable context selector cannot receive the true sense label at inference. Let learned router logits $\ell_{\theta,s}(x,t)$ depend only on the available sequence and token state, and define

$$p_\theta(s \mid x,t) = \frac{\exp(\ell_{\theta,s}(x,t)/\tau_{\text{r}})}{\sum_{s'} \exp(\ell_{\theta,s'}(x,t)/\tau_{\text{r}})}, \qquad \tau_{\text{r}} > 0. \tag{47}$$

The router may use the illustrative score in Eq. (43), but need not do so. Its weights define two possible shared-sector filters:

$$\begin{aligned} \bar{h}_t^{(\Pi)}(x) &= \sum_s p_\theta(s \mid x,t)\, \Pi_{t,s} h(x), \\ \bar{h}_t^{(A)}(x) &= \sum_s p_\theta(s \mid x,t)\, A_{t,s} h(x). \end{aligned} \tag{48}$$

Hard selection uses the largest router probability; soft selection uses Eq. (48). The effective operator $\sum_s p_\theta(s \mid x,t)\Pi_{t,s}$ is generally not idempotent and hence is not itself an orthogonal projector. Attention may supply its routing signal, but neither the shared sector nor the operator structure follows from standard attention. The constrained model must therefore be compared with parameter-matched mixture-of-experts and unstructured low-rank alternatives.

For an implementable adapter at a chosen layer, let $Z_{t,s} \in \mathbb{R}^{d\times m_s}$ be a trainable seed matrix such that $(I-P_t)Z_{t,s}$ has full column rank, and set $B_{t,s} = \mathrm{orth}((I-P_t)Z_{t,s})$. Tie the same Q_t across all s, compute the router, and choose $\bar{h}_t \in \{\bar{h}_t^{(\Pi)}, \bar{h}_t^{(A)}\}$ from Eq. (48). The filtered state may be used as a diagnostic probe or inserted through the residual update $h^+ = h + \eta U\bar{h}_t$, where $U \in \mathbb{R}^{d\times d}$ is a learned output map and η is a scalar gate. The router, bases, spectra, and adapter are then trained on a language-model or task loss; parameter tying enforces the shared sector and the factor $(I - P_t)$ enforces transversality. Held-out evaluation must compare against adapters with the same ranks and parameter count but no shared-sector constraint.

VII. DISCUSSION

The geometric argument now has four deliberately separated layers. Lévy's lemma explains typical near-orthogonality on the sphere; JL explains why a finite configuration of exponentially many directions can retain its metric relations in logarithmic dimension; random spherical coding calibrates the largest overlap; and active-set analysis asks what a decoder can recover. Each ingredient is standard, but their conjunction exposes the central tension: quasi-orthogonal code cardinality can grow exponentially while angular resolution and simultaneous accessibility remain finite. The discriminating LLM question is whether training redistributes the unavoidable overlap so that frequently coactive features incur less held-out interference than isotropic and activity-matched controls.

The shared-sector operator proposal is logically independent of that null model. It asserts that trajectories of some lexically ambiguous tokens admit a specific constrained organization. Neither a fitted common subspace nor a nonzero commutator is sufficient evidence. Evidence requires generalization to held-out contexts and an advantage over equally expressive unstructured projectors or mixture models.

A. Relation to superposition and sparse autoencoders

The superposition hypothesis holds that neural networks may represent more features than they have neurons by using nonorthogonal directions with sparse activity [4, 26]. The null model separates dictionary coexistence from simultaneous linear readout. In the isotropic, weak-dependence model, many directions can coexist while a fixed-target correlation decoder accumulates interference of scale $\sqrt{k/d}$. Sparse activity controls that interference for this decoder; it is not asserted to be universally necessary for structured dictionaries or nonlinear recovery.

Sparse autoencoders (SAEs) give a concrete empirical interface. They approximate layer activations as sparse expansions over learned decoder directions, making the w_i of Eq. (22) measurable objects rather than hypothetical features. Recent work has scaled SAEs, studied SAE feature geometry, and automated interpretation of large numbers of latents [27–30]. In the present framework, SAE decoder directions make the Lévy/JL calibration and coactivation-weighted diagnostic measurable, while sense-conditioned SAE activation clusters can test the stronger reference-subspace hypothesis.

B. Limitations

The framework can fail in several ways.

First, Lévy's lemma controls a regular probe or a fixed overlap; mutual quasi-orthogonality of a finite family additionally requires a union bound or code construction. JL supplies a constructive exponential scale, not a hard upper bound on spherical codes. The deterministic Welch bound constrains a specified M-vector dictionary, while Eq. (17) describes an independent isotropic ensemble. None is a semantic impossibility theorem.

Second, the double exponential is conditional on fixed $q > 1$, fixed $0 < \varepsilon < 1$, and access to a full carrier with tensor dimension $d = q^n$. Sequence length, compositional language, or the existence of many possible contexts does not by itself give a Transformer residual-stream tensor product. Restriction to product vectors or another low-complexity family may remove the second amplification. Without the full-carrier premise, the ambient code has an existence lower bound exponential, not doubly exponential, in measured d.

Third, the nominal layer width need not equal the selected carrier rank d. Contextual embeddings can be anisotropic, and the occupied carrier can vary strongly by layer and training regime [31–34]. Whitening and related postprocessing can partially restore isotropic geometry [32, 33, 35]. At the same time, anisotropy should not be treated as an automatic or universal property of all Transformer representations [36]. The rule used to select or whiten a d-dimensional carrier must therefore be reported. Whitening also changes the metric in which overlaps are measured, and the covariance spectrum may contain information not summarized by one rank. We do not introduce a second dimension symbol, but d is protocol-dependent rather than an intrinsic semantic constant.

Fourth, M depends on the feature-extraction, normalization, and duplicate-removal rules. A claimed departure from the null may instead expose a poor dictionary, dependence among samples, finite-size effects, or a changed metric. These alternatives must be checked before attributing the departure to learned semantic organization.

Fifth, the reference-subspace pattern may simply be descriptively wrong. Trained models may represent lexical ambiguity without a preferred token-associated direction or subspace. A low-rank fit found after inspecting test labels is not evidence; ranks, candidate references, and routers must be selected on training data and evaluated out of sample.

Sixth, the law $\sigma_{\rm int} \sim \sqrt{k/d}$ concerns a fixed-target correlation readout with random-like directions and weakly correlated signs or amplitudes. It is neither a joint sparse-recovery theorem nor a universal equality. Structured dictionaries, nonlinear decoders, multiple comparisons over inactive features, and correlated errors can change the scaling.

The coactivation energy in Eq. (29) is likewise a diagonal proxy. If interference contributions from distinct features are correlated, the full importance-weighted squared error contains additional covariance terms and must be estimated directly.

Finally, the operator model has choices: the candidate reference subspace $\mathcal{R}_t$, the router, the local ranks, and any spectra in Eq. (38). The true sense label cannot be supplied to the router at test time. Eigenvalues are not identified by subspaces alone, and a fitted nonzero commutator is generic. These choices add evidence only through held-out prediction, compression, or prespecified order effects.

C. Empirical predictions

The key empirical task is to distinguish the applicability of the Lévy/JL geometry from three stronger semantic hypotheses.

Geometric calibration (not a semantic claim). For a dictionary independently established to be overcomplete, compare its coherence, overlap distribution, and fixed-target readout error with isotropic and activity-matched controls. Agreement with Eq. (17) is a baseline result, not evidence that overcompleteness is useful; departure from it must be diagnosed before being called learned structure. JL itself is a theorem; the empirical question is whether task-relevant geometry and readout survive projection near its logarithmic dimension scale.

Claim A: privileged token reference subspace. For a lexically ambiguous token t, there exists a privileged token-associated subspace $\mathcal{R}_t$ such that sense-relevant variation is better exposed in $\mathcal{R}_t^{\perp}$ than in the orthogonal complements of matched control subspaces of the same dimension.

Claim B: low-dimensional sense subspaces. Within $\mathcal{R}_t^{\perp}$, sense-conditioned states admit a cross-validated low-rank description that predicts or compresses held-out states better than shuffled-label, ordinary clustering, and matched-subspace controls. Related polysemous senses need not have minimal cross-sense coherence.

Claim C: shared-sector class operators. A non-oracular router and class-conditioned projectors $\Pi_{t,s}$ sharing P_t improve held-out contextual prediction or compression relative to parameter-matched unstructured projectors and mixture-of-experts models. A commutator matters only if it predicts order effects defined independently of the fitted operators.

These claims can fail independently. The geometric calibration can hold while all three structural claims fail. Claim A can hold while sense information remains high-dimensional. Claim B can hold relative to a non-token-associated subspace, in which case Claim A fails. Claims A and B can hold even if the operator constraint in Claim C adds no predictive value.

The following tests separate the possibilities.

1. *Lévy/random-code calibration.* Fix the carrier metric, spectral threshold or whitening rule, feature extractor, and duplicate criterion before evaluation. Report the selected rank d, dictionary size M, maximum coherence, the full overlap distribution, and active-set statistics. Compare with the Welch floor, isotropic dictionaries, and controls matched for marginal activity and coactivation. Treat Eq. (17) as a finite-sample null prediction, not a pass/fail capacity test. Also compare the signed pairwise distribution with the independently isotropic variance $1/d$; deviations may reflect dependence, anisotropy, extraction choices, or learned alignment.

2. *JL projection-stability test.* Fix a finite evaluation set of M normalized directions or contextual states, include the origin, and prespecify a relative squared-distance distortion $\delta_{\rm JL}$. Project to $d_{\rm proj}$ dimensions using frozen random maps and measure pairwise distortion and held-out task performance as $d_{\rm proj}$ varies. JL predicts metric preservation on the scale $d_{\rm proj} = O(\delta_{\rm JL}^{-2}\log(M+1))$; preservation of semantic performance is the additional empirical question. Compare random Gaussian or orthogonal maps with learned low-rank maps at the same $d_{\rm proj}$.

3. *Reference-subspace test.* For a lexically ambiguous token, choose candidate reference subspaces generated by the mean contextual embedding, a static input embedding explicitly mapped into the same layer carrier, learned token-specific directions, or a low-dimensional combination of these. Project layer states as in Eq. (45). Sense labels should be better separated in $\mathcal{R}_t^{\perp}$ than in orthogonal complements obtained after ordinary centering, token-mean subtraction, top-principal-component removal, and random, frequency-matched, or layer-matched control subspaces of the same rank. Compare held-out classification, compression, or incremental mutual information, with permutation over sense labels and bootstrap over token instances.

4. *Operational subspace test.* For each sense s, collect projected states $z_\ell(t,x) \in \mathcal{R}_t^{\perp}$ and estimate a low-rank subspace by PCA or probabilistic factor analysis. Select m_s without test-label leakage and use an

orthonormal basis $B_{t,s}$. Evaluate held-out reconstruction and sense prediction against shuffled labels, ordinary supervised subspaces, and unsupervised clustering of equal rank. Report $\|B_{t,s}^{\mathsf{T}} B_{t,s'}\|_{\mathrm{op}}$ descriptively, without assuming related senses must be nearly orthogonal.

5. *Reference-versus-sense readout test.* Linear probes trained on $z_\ell(t,x)$ should perform comparably to probes trained on the full state $h_\ell(t \mid x)$, while probes restricted to the reference component $P_t h_\ell(t \mid x)$ should provide less incremental sense information after frequency, syntax, and sense-prior controls. Near-chance performance is not required because the reference component may legitimately retain correlated information.

6. *Shared-sector operator test.* Fit P_t, $B_{t,s}$, $A_{t,s}$, and the router on training sequences, then evaluate Eq. (48) without true sense labels on held-out instances. Compare with unconstrained low-rank projectors, ordinary contextual probes, parameter-matched mixture-of-experts models, and shuffled class or sense labels. If an order claim is made, define the sequential linguistic or model intervention independently; then ask whether $\|[\Pi_{t,s}, \Pi_{t,s'}]\|_{\mathrm{op}}$ predicts the measured asymmetry. Merely obtaining a nonzero commutator does not pass the test.

7. *Synthetic concurrency degradation.* At a fixed layer, inject controlled superpositions

$$h_k = h_0 + \lambda \sum_{j=1}^{k} x_j w_j, \tag{49}$$

where h_0 is a held-out baseline activation, the w_j are independently chosen unit injection directions, the coefficients x_j are independent, centered, and unit-variance, and λ controls injection amplitude. The directions may come, for example, from SAE decoder directions or frozen linear probes. As k increases, readout should degrade smoothly on the scale $\sqrt{k/d}$ for random-like directions under the specified correlation decoder. Include norm-preserving, random-direction, activation-frequency, and out-of-distribution controls; degradation by itself does not validate semantic superposition.

8. *Coactivation-dependent geometry.* At matched marginal activation frequency, feature similarity, SAE reconstruction error, amplitude, and task relevance, important frequently coactive features should contribute less to Eq. (29) than activity-matched controls. This tests learned redistribution of interference, independently of the reference-subspace hypothesis.

D. Scope of the quantum analogy

The analogy to quantum contextuality is structural and limited. In the proposed construction, every $\Pi_{t,s}$ acts identically on the invariant subspace $\mathcal{R}_t$. Under the spectral-separation condition stated after Eq. (38), the observables $A_{t,s}$ additionally share P_t as the spectral projector of λ_t. The shared term cancels from the commutator, so it is the relative geometry of the sense sectors that may produce order dependence. Figure 1 records the displayed incidence pattern; omitted complementary outcome sectors would be required to complete each measurement context in a full contextuality hypergraph. Equations (36)– (38) provide the corresponding operator parameterization. Vectors, subspaces, projectors, and observables are distinct mathematical types.

Quantum contextuality has the force of no-go theorems such as Kochen–Specker [37, 38]; no analogous theorem is claimed here for semantic embeddings. Real symmetric matrices can be noncommuting while still being stored and evaluated on an ordinary classical computer, and generic fitted projectors are commonly noncommuting. Thus noncommutativity by itself supplies neither a semantic result, Born probabilities, nor physical quantumness.

Three levels of analysis should not be conflated.

1. *Classical high-dimensional representation.* A present-day Transformer carries real-valued hidden states through attention, nonlinear feed-forward maps, residual connections, and normalization. Concentration, quasi-orthogonality, feature superposition, dynamic embeddings, and finite readout are all available at this level. None requires quantum probability or quantum hardware.

2. *Quantum-like semantic or contextual formalism.* States, contexts, bases, interference terms, noncommuting questions, or measurement-like updates may be used as an abstract model of meaning even when the implementation remains classical. The relevant issue is not the vocabulary but whether this extra structure improves compression, explanation, or prediction relative to an ordinary contextual embedding model.

3. *Genuine quantum machine learning or quantum attention.* Here semantic states would be encoded in qubits or other physical quantum degrees of freedom, intermediate evolution would be unitary, phase would affect interference, and measurement would follow quantum probability. A standard Transformer does none of these things.

The random-dictionary and linear-readout baselines belong entirely to the first level. Shared-sector projection is also an ordinary classical computation. The proposal earns the stronger quantum-like interpretation only if its operator constraints or prespecified order effects improve prediction, compression, or explanation relative to equally expressive classical baselines. Even a positive result would support a useful formalism, not physical quantum computation; phase-sensitive dynamics and Born-rule measurement remain absent.

The conditional double exponential has its most direct realization in a full tensor-product Hilbert space, where n local q-level factors give $d = q^n$ before quasi-orthogonal packing is considered. This is a statement about composition and finite angular resolution, not a signature of physical quantumness; classical tensor-product feature spaces can exhibit the same counting. Conversely, importing $d = q^n$ into an ordinary LLM merely because language is compositional would be unwarranted. The carrier factorization and access to the full carrier, rather than only product states or another low-complexity manifold, must be demonstrated.

The dimension of $\mathcal{R}_t$ is also explicit. In Hilbert-space quantum logic, two or more contexts may share one spectral sector, or several; a four-dimensional example has two contexts with two common observables [23]. The semantic analogue is $\dim(\mathcal{R}_t) = p$: $p = 1$ is the simplest case, while $p > 1$ permits several token-associated reference directions. In every case, sense-specific information is hypothesized to occupy subspaces of $\mathcal{R}_t^{\perp}$.

VIII. CONCLUSION

The central geometric conclusion is an abundance–resolution tradeoff. Lévy's lemma says that regular probes concentrate on a high-dimensional sphere and, in particular, that almost every direction has small overlap with any fixed direction. Johnson–Lindenstrauss supplies the complementary finite-set statement: logarithmic exact dimension suffices to retain the pairwise geometry of many points. Applied constructively to an orthogonal family, it yields quasi-orthogonal codes of size $M_\varepsilon(d) \geq \exp(c\varepsilon^2 d)$ for fixed tolerance and sufficiently large d. Direct random spherical coding recovers the same exponential scale, and the exact complex-Haar law gives Eq. (19), whereas the Welch bound gives the hard coherence floor for any specified overcomplete family. These are distinct views—typicality, finite-set embedding, random construction, and deterministic obstruction—of the same quasi-dimensional geometry.

A genuine double exponential requires two amplifications. If an independently justified, fully accessible tensor-product carrier gives $d = q^n$, then for fixed $q > 1$ and fixed $0 < \varepsilon < 1$ the quasi-orthogonal code cardinality has an existence lower bound of the form $\exp(c\varepsilon^2 q^n)$, doubly exponential in the factor count n. The claim need not survive restriction to product vectors or another low-complexity state family. Nothing in Lévy's or the JL lemma alone makes the number of semantic directions doubly exponential in an ordinary LLM width. For a Transformer residual stream treated simply as $\mathbb{R}^d$, the supported conclusion is an ambient-code existence lower bound exponential in measured d; tensor-product growth must be established separately.

The same high-dimensional geometry that permits large codes also concentrates typical angles and limits access. In the small-overlap regime $\log M \ll d$, an independently sampled isotropic dictionary has a largest-overlap scale of order $2\sqrt{\log M/d}$, and a random-like active set produces fixed-target correlation-readout interference of order $\sqrt{k/d}$. The substantive semantic hypothesis concerns where the remaining overlap is placed: important features that frequently coactivate should contribute less coactivation-weighted interference than activity-, amplitude-, and importance-matched controls. This connects the Lévy/JL code-cardinality picture to observable readout rather than equating code cardinality with usable meaning.

The second proposal is a constrained operator model of lexical ambiguity. Class-conditioned projectors share a token-associated invariant sector and differ on low-dimensional sense subspaces in its orthogonal complement. Conditional projection or observable filtering yields a class-conditioned embedding, and a learned non-oracular router selects or mixes classes from the observed context. The common projector does not cause noncommutativity, and a nonzero fitted commutator has no evidential force by itself. The model succeeds only if the shared-sector constraint improves held-out prediction or compression over parameter-matched unstructured models, or if its commutators predict independently specified order effects.

The geometric calibration, coactivation hypothesis, and shared-sector model can fail independently. The construction is quantum-inspired, not a claim that a Transformer is a quantum device: it uses real matrices on classical hardware and introduces no Born rule, unitary dynamics, or physical measurement. Quantum terminology is earned only if the constrained operator structure provides additional held-out explanatory or predictive value.

Appendix A: Minimal formalization of the three computational loops

The following subsections correspond one-to-one to Secs. II A–II C. Let Θ collect all persistent real-valued model parameters, let $\mathcal{V}$ be the token vocabulary, and let Tok and Detok denote tokenization and detokenization. Hidden states lie in the finite real inner-product space $\mathbb{R}^d$.

1. Creating the weights

For a training string σ, write

$$x^{(\sigma)}_{1:T_\sigma} = \mathsf{Tok}(\sigma), \qquad e^{(\sigma)}_i = E_{\mathrm{tok}}[x^{(\sigma)}_i] \in \mathbb{R}^d,$$
$$h^{(L,\sigma)}_{1:T_\sigma} = F_\Theta\left(e^{(\sigma)}_{1:T_\sigma}; 1{:}T_\sigma\right). \tag{A1}$$

Here E_{tok} is the learned embedding table, and $1{:}T_\sigma$ records that the forward map receives positional information. The latter may be additive, rotary, or relative. The map F_Θ contains causal self-attention of the form already displayed in Eq. (46), as well as residual, normalization, multi-head, and feed-forward operations. During this forward pass Θ is fixed. The attention coefficients and hidden states computed by F_Θ are transient.

With vocabulary logits $g_{\sigma,i} = W_{\text{out}} h_i^{(L,\sigma)} + b$, next-token training may be written schematically as

$$\widehat{\mathcal{L}}_{\mathcal{B}_r}(\Theta) = -\sum_{\sigma\in\mathcal{B}_r}\sum_{i=1}^{T_\sigma-1}\log\Big[\operatorname{softmax}(g_{\sigma,i})_{x_{i+1}^{(\sigma)}}\Big],$$
$$\Theta^{(r+1)} = \operatorname{Opt}\Big(\Theta^{(r)}, \nabla_\Theta \widehat{\mathcal{L}}_{\mathcal{B}_r}(\Theta^{(r)})\Big). \tag{A2}$$

Here $\mathcal{B}_r$ is the minibatch at optimization step r. Equation (A1) changes activations during a forward pass; Eq. (A2) changes persistent weights between passes.

2. Generating the output

Let $u_{1:m} = \mathsf{Tok}(\sigma_{\text{prompt}})$ and let $\Theta^\star$ denote the fixed deployed parameters. At generation step $j \ge m$, the forward map returns logits $g_j(\Theta^\star; u_{1:j})$, and for $\tau_{\text{dec}} > 0$ a temperature-scaled sampling distribution is

$$\pi_j(a) = \frac{\exp[g_{j,a}(\Theta^\star; u_{1:j})/\tau_{\text{dec}}]}{\sum_{b\in\mathcal{V}}\exp[g_{j,b}(\Theta^\star; u_{1:j})/\tau_{\text{dec}}]}, \qquad a \in \mathcal{V},$$
$$u_{j+1} \sim \pi_j,$$
$$u_{1:j+1} = (u_{1:j}, u_{j+1}). \tag{A3}$$

A greedy or truncated decoding rule may replace direct sampling. If generation stops at J, the returned string is $\widehat{s} = \mathsf{Detok}(u_{m+1:J})$. The prefix and all context-dependent states change with j, but $\Theta^\star$ does not. A key–value cache stores previously computed activations and likewise leaves $\Theta^\star$ unchanged.

3. Recursive adaptation, alignment, and hallucination

Post-training alignment first produces a deployed starting point $\Theta_{\text{align}} := \Theta^{(0)} = \operatorname{Align}(\Theta_{\text{base}}; \mathcal{D}_{\text{inst}}, \mathcal{D}_{\text{pref}})$ from instruction and preference data. Genuine online adaptation adds a parameter update after interaction round r:

$$\Theta^{(r+1)} = \Theta^{(r)} - \eta_r \mathcal{P}_r^{(\Theta)} \nabla_\Theta \mathcal{J}_r(\Theta^{(r)}; \mathcal{B}_r), \qquad \eta_r > 0. \tag{A4}$$

Here $\mathcal{B}_r$ is verified adaptation data, $\mathcal{J}_r$ is its declared adaptation objective, and $\mathcal{P}_r^{(\Theta)}$ is a mask or projection in parameter space. Setting $\mathcal{P}_r^{(\Theta)} = 0$ recovers ordinary frozen-weight inference; restricting it to adapter coordinates confines the update to those parameters; the identity permits a full update. Boundedness additionally requires clipping, a trust region, or an explicit norm bound. An external-memory update instead has $m_{r+1} = U_{\text{mem}}(m_r, \iota_r)$, where the interaction record ι_r may contain the round-r input, response, tool results, and verified feedback. It leaves Θ fixed and is therefore not Eq. (A4).

Finally, let $H(\Theta; \mathcal{Q}, \mathcal{G}, \mathcal{D})$ be a prespecified held-out rate of unsupported claims for prompt distribution $\mathcal{Q}$, grounding condition $\mathcal{G}$, and decoding rule $\mathcal{D}$. The causal contrast requested in Sec. II C is

$$\Delta_{\text{hall}}^{\text{align}} = H(\Theta_{\text{align}}; \mathcal{Q}, \mathcal{G}, \mathcal{D}) - H(\Theta_{\text{ctrl}}; \mathcal{Q}, \mathcal{G}, \mathcal{D}). \tag{A5}$$

The control Θ_{ctrl} starts from the same base and is matched in data exposure, optimization and compute budget, and random seeds, except for the declared alignment intervention. One may set $\Theta_{\text{ctrl}} = \Theta_{\text{base}}$ only when "no post-training" is the explicit control condition. Coverage and abstention must be reported beside H, since universal refusal could otherwise lower the unsupported-claim rate without adding knowledge.

The hypothesis that alignment increases hallucination in the declared regime is $\Delta_{\text{hall}}^{\text{align}} > 0$; reduction corresponds to a negative value. Nothing in the formalism fixes the sign. A cumulative online update is tested against a frozen control exposed to the same interaction stream, not merely against its own earlier parameter snapshot.

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